

Martin Lavin Carvajal

Santiago, Chile | martinlavinc@gmail.com | +56 9 4084 0807
linkedin.com/in/martin-lavin-carvajal-010b08339 | github.com/mlavinc
Portfolio: <https://mlavinc-portfolio.vercel.app/>

PROFESSIONAL PROFILE

Final-year Computer and Informatics Civil Engineering student with hands-on experience in process automation and the development of technology solutions within corporate environments. Engineering background with a strong foundation in mathematics, physics, and project management. Interested in Cloud Engineering, full stack development, distributed systems architecture, and FinOps, with a focus on building scalable solutions.

EDUCATION

Universidad Mayor — Santiago, Chile 2022 – 2026
B.Eng. in Computer and Informatics Civil Engineering

WORK EXPERIENCE

Nestlé Digital & New Tech 2025
Professional Internship — Computer Engineering

- Participated in regional digitalization and automation initiatives using Power Platform and Python.
- Developed an automation solution for the Annual Performance Evaluation process, adopted by employees nationwide.
- Implemented a digital onboarding system for a team of approximately 20 people in the NDC area.
- Developed an SH&E registration system for access control at the head office, featuring ID (RUT) verification and induction history tracking.
- Automated the power-tool registration process for the technical plant, replacing a manual, paper-based workflow.
- Provided IT support and incident resolution at the head office.

Universidad Mayor 2023
Teaching Assistant — Mathematics Department

- Provided academic support in Differential Calculus, Integral Calculus, Algebra, Geometry, and Differential Equations.
- Resolved student questions and provided guidance throughout the learning process, reinforcing analytical skills.

SIMCE (Chilean Ministry of Education) 2024 – 2025
Standardized Test Examiner

- Administered standardized assessments across primary and secondary education institutions.
- Enforced testing protocols and managed group logistics during administration sessions.

PERSONAL PROJECTS

Skill Tracker: Full-stack web application for tracking skills and competitive rankings, built with a REST API architecture and deployed via Docker containers.

Stack: Go, React, PostgreSQL, Docker

<https://github.com/mlavinc/skill-tracker>

Electrocardiogram AI Serverless: ECG signal classifier based on Random Forest, deployed as a serverless architecture on AWS Lambda with API Gateway.

Stack: Python, Scikit-learn, NumPy, AWS Lambda, S3, API Gateway

<https://github.com/mlavinc/ecg-ai-serverless>

Cloud Operations Lab: Cloud operations lab covering infrastructure as code, Linux server administration, and operational process automation.

Stack: AWS (EC2, IAM, VPC, CloudWatch, SSM), Terraform, Bash, Python

<https://github.com/mlavinc/cloud-operations-lab>

Document RAG Knowledge Agent: Full-stack RAG system for natural language document querying, built with a microservices architecture deployed on AWS and semantic search using vector embeddings.

Stack: React/TypeScript, Node.js/Express, FastAPI, AWS Bedrock (Titan + Claude Haiku), Aurora pgvector, Lambda, S3, CloudFront

<https://github.com/mlavinc/document-knowledge-agent>

ThermoSim: Interactive engine for modeling and solving Thermodynamics problems, including open and closed systems, energy balances, thermodynamic properties, heat transfer, and entropy.

Stack: Go, Cobra, Viper, Bubble Tea

<https://github.com/mlavinc/ThermoSim>

Thesis Project (in progress): Research on AI applied to medical imaging, focused on models robust to domain shift, combining radiomics, fractal biomarkers, and Deep Learning with a cloud-based architecture. Developed in collaboration with a PhD researcher in Medical Imaging.

CERTIFICATIONS

- Google Cloud Academy — Feature Engineering on Google Cloud (2025)
- Google Cloud Academy — Build, Train and Deploy ML Models with Keras on Google Cloud (2025)
- Cisco Networking Academy — Networking Basics
- Cisco Networking Academy — Introduction to Cybersecurity and Ethical Hacking
- AWS Cloud Practitioner Essentials

TECHNICAL SKILLS

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, VPC, CloudWatch, Cost Explorer), GCP (BigQuery), Terraform, Docker, Linux, Bash

Architecture & Distributed Systems: Microservices, Event-Driven Architecture (EDA), Fan-Out/Pub-Sub patterns, Transactional Outbox Pattern, idempotency, message brokers, API versioning

Programming Languages: Python, Go, Java, C#, C++

Databases: SQL, BigQuery

Machine Learning / AI: Scikit-learn, TensorFlow, OpenCV

AI-Assisted Development Tools: Claude Code, Cursor AI

Automation: Power Apps, Power Automate

Version Control: Git, GitHub

ADDITIONAL INFORMATION

Languages: Spanish (native), English (C1, Cambridge)